

Group name

Enter your details :

Section

Instructions/information: Following are some exercises that relate to a reading you have been assigned. Please refer to your notes as you answer them. You are encouraged to confer with your group mates as you put down your thoughts.

4.1 The Idea of Intervention

4.1.1 Module 5 | Interventions as Actions

Reflect and discuss (A) Interventions as Actions 1

 1. Which of the following are actions _____? (choose all that apply)

- A. Turning off the computer at which you are working.
- B. Removing corruption from politics.
- C. Not getting a cold.
- D. Taking 200% of the daily allowance of vitamin C.

 2. Which of the following are practical *or* hypothetical _____, as opposed to actions that are too vague to discuss sensibly? (choose all that apply)

- A. Making all residents of New Jersey move to Nevada.
- B. Treating men and women the same.
- C. Eliminating racism.
- D. Putting a full time police officer in charge of guarding every apartment building in NYC.

4.1.2 Module 5 | Ideal Interventions

Reflect and discuss (B) Ideal Interventions 1

 3. Consider a system describing an airplane flying at cruising altitude with two variables:

- A. Seat Belt Sign [lit, off]
- B. Turbulence [yes, no]

If the pilot hears a weather report from a plane flying 100 miles ahead that turbulence is coming, and hits a switch to light up the Seat Belt Sign, this is not _____ on the Seat Belt Sign. That is because: (choose one)

- A. The intervention is not effective at determining the value of the variable it targets.
- B. The intervention does not directly target only the seat belt sign.
- C. The intervention is not exogenous to the system, it has been caused by some other variable, in this case a weather report of turbulence.

Reflect and discuss (C) Ideal Interventions 2

 4. Which of the following interventions would be _____ at determining the value of the variable: Computer is turned on and functioning properly [yes, no]? (choose all that apply)

- A. Sitting down for one hour and trying to make a computer function perfectly.
- B. Copying a file onto a computer that is infected with a computer virus.
- C. Removing the power source from a computer.
- D. Beating a computer with a sledgehammer for an hour.

 5. Suppose the causal system under consideration involves three variables: {Foot Speed, Endurance, Confidence}. Which of the following interventions are directly targeted only at _____, and are not “fat hand” interventions? (choose all that apply)

- A. Hypnotizing the athlete into believing that he is great.
- B. Administering a drug that relieves anxiety but also relaxes muscles.
- C. Having the athlete watch videos of all his past performances that were successful.
- D. None of the above.

4.2 Interventions and Causal Relationships

4.2.1 Module 5 | Intervening on Causes

Reflect and discuss (D) Intervening on Causes 1

 6. Let's assume that high cholesterol causes poor artery condition. If we ideally intervene to change your _____, then...(choose one)

- A. the relationship between Cholesterol Level and Artery Condition changes.
- B. ...the value of Cholesterol Level is determined by the intervention.
- C. ...the relationship between Cholesterol level and Heart Condition changes.
- D. None of the above.

 7. In general, an _____ on a cause...

- A. determines the value of the cause.
- B. changes the relationship between the cause and its effects.
- C. always changes the value of the cause.
- D. None of the above.

4.2.2 Module 5 | Intervening on Effects

Reflect and discuss (E) Intervening on Effects 1

 8. Let's assume that _____ causes poor artery condition. If we ideally intervene on the condition of your arteries, then...(choose all that apply)

- A. the relationship between Cholesterol Level and Artery Condition changes, at least temporarily.
- B. the value of Cholesterol Level changes.

C. the value of Artery Condition is always changed.

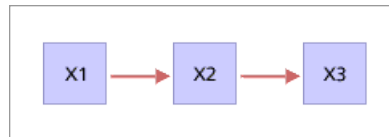
9. Assuming that there is _____ between causes and effects, then in general, an ideal intervention on an effect...(choose all that apply)

- A. does nothing to the value of the causes of the effect.
- B. changes the relationship between the effect and its causes.
- C. None of the above.

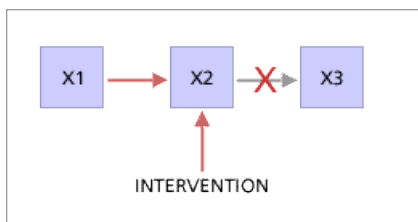
4.2.3 Module 5 | Natural and Experimental Causal Systems

Reflect and discuss (F) Natural and Experimental Causal Systems 1

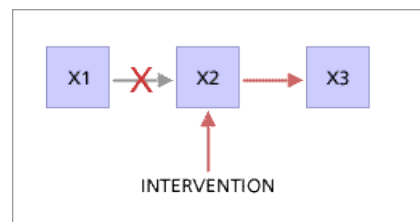
10. Suppose that the following causal graph represents the pre-manipulated causal system governing the variables X1, X2, and X3.



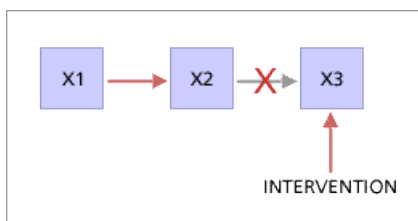
Which of the following causal graphs represents the post-manipulation system following an ideal intervention on _____?



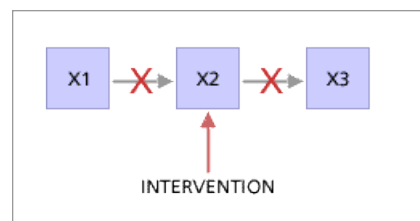
A.



B.

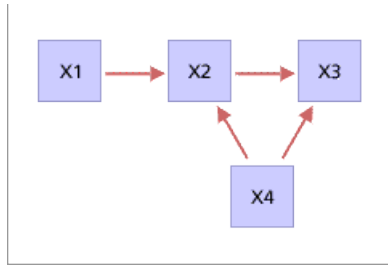


C.

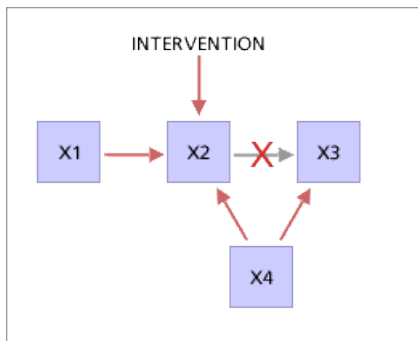


D.

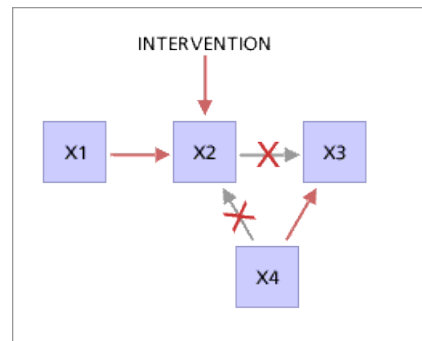
11. Suppose that the following causal graph represents the pre-manipulated causal system governing the variables X1, X2, X3, and X4.



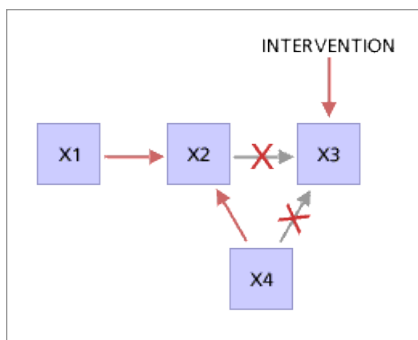
Which of the following causal graphs represents the post-manipulation system following an ideal intervention on _____?



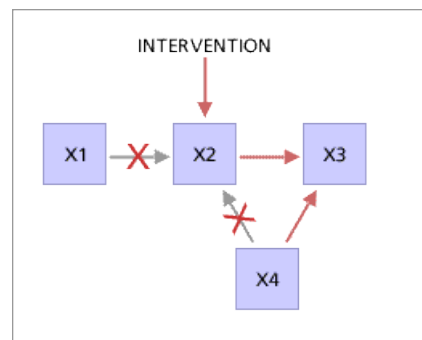
A.



B.



C.



D.

4.2.4 Module 5 | Exogenous and Endogenous Variables and Causal Structure

🧠 Reflect and discuss (G) Exogenous and Endogenous Variables 1

📝 12. Consider the following causal system representing a bicycle: {Gear selected, Pedals moving} Is the variable _____ dependent, or independent? (choose one)

- A. Independent
- B. Dependent

📝 13. Consider the following causal system describing the causal relations inside an automobile: {Gear shift

setting, Accelerator pedal setting, Engine RPMs}. Is the variable _____ dependent or independent? (choose one)

- A. Dependent
- B. Independent

 14. Consider the solar system, with variables 'distance of the Earth from the sun', 'temperature of the sun', and 'temperature of the Earth'. Is the variable _____ dependent, or independent? (choose one)

- A. Dependent
- B. Independent

 15. Consider a following system: a human body, with variables 'staph infection', 'fever', and 'body temperature'. Is the variable _____ dependent or independent? (choose one)

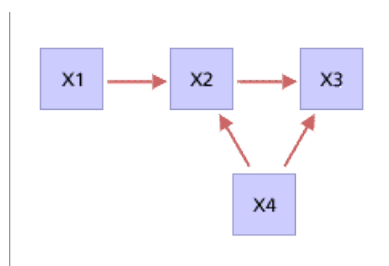
- A. Dependent
- B. Independent

 16. Consider the electrical system of a house, which includes variables {Lamp [on, off], Refrigerator [on, off], Radio [on, off], Lamp Switch [up, down]}. Is the variable _____ dependent, or independent? (choose one)

- A. Independent
- B. Dependent

Reflect and discuss (H) Exogenous and Endogenous Variables 2

 17. Use this graph to answer the following question:



For which of the following variables would an intervention on it be _____? (choose all that apply)

- A. X1.
- B. X2.
- C. X3.
- D. X4.

4.2.5 Module 5 | Interventions on Cyclic Causal Systems

Reflect and discuss (I) Interventions on a Cyclic System 1

 18. Consider the causal system of the bicycle in the simulation to have three variables:

- A. Pedals [moving, still]
- B. Rear Wheel [moving, still]
- C. Gear Shift [forward, back]

Choose all of the following that are _____ in this system.

- A. Pedal $\rightarrow$ Rear Wheel.
- B. Rear Wheel $\rightarrow$ Pedal.
- C. Pedal $\rightarrow$ Gear Shift.
- D. Gear Shift $\rightarrow$ Pedal.
- E. Rear Wheel $\rightarrow$ Gear shift.
- F. Gear Shift $\rightarrow$ Rear Wheel.

Reflect and discuss (J) Interventions on Cyclic Systems 2

 19. Consider the causal system of the bicycle in the second simulation to have three variables:

- Pedals [moving, still]
- Rear Wheel [moving, still]
- Gear Shift [forward, back]

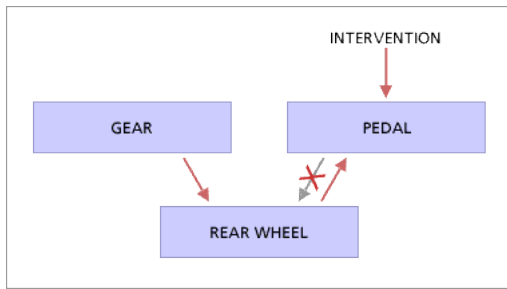
Choose all of the following that are _____ in this system.

- A. Pedal $\rightarrow$ Rear Wheel.
- B. Rear Wheel $\rightarrow$ Pedal.
- C. Pedal $\rightarrow$ Gear Shift.
- D. Gear Shift $\rightarrow$ Pedal.
- E. Rear Wheel $\rightarrow$ Gear shift.
- F. Gear Shift $\rightarrow$ Rear Wheel.

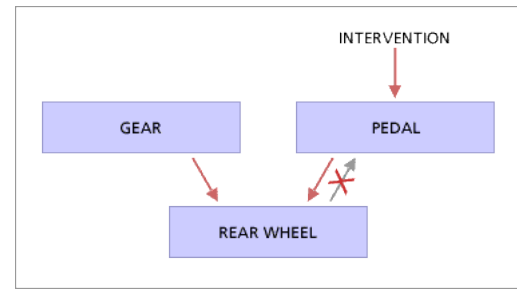
Reflect and discuss (K) Interventions on Cyclic Systems 3

 20. Which of the following is the post-manipulation graph for an intervention on _____?
(choose one)

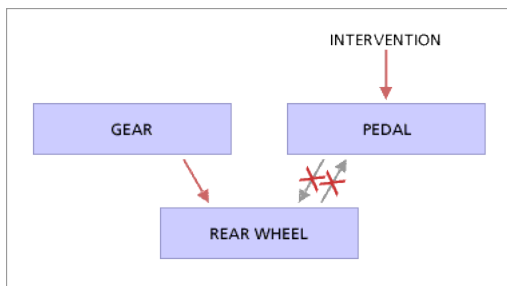
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A.



B.



C.

None of these.

D.

4.2.6 Module 5 | Case Studies

Reflect and discuss (L) Case Studies 1

21. Which of the following variables was most directly _____ the city officials of Cali and Bogotá? (choose one)

- A. Homicide Rate
- B. Number of People Carrying Guns
- C. Crime Rate
- D. Frequency of Gun Searches

22. Relative to the following variables:

- Frequency of Searches
- Homicide Rate
- Number of Police Patrolling
- Number of People Carrying Guns

The gun ban would be _____ if:

- A. It increased the number of homicides
- B. It directly increased both the frequency of gun searches and the number of police patrolling the streets
- C. It reduced the number of people carrying guns but not the number of homicides
- D. It reduced the number of people carrying guns, which reduced number of homicides.

 Reflect and discuss (M) Case Studies 2

 23. Which of the following variables was most directly _____ of the researchers in the study described in the article? (choose one)

- A. Participation in TV and Video-Game Reduction Lessons
- B. Number of Hours Spent Playing Video Games
- C. Diet
- D. Number of Hours Spent Watching TV

 24. Consider the following variables:

- Time Spent Watching TV
- Sensitivity towards other children
- School Performance

Participation in the TV reduction lessons would be _____ if:

- A. It directly affects both time spent watching TV and sensitivity towards other children
- B. It directly decreases the amount of time watching TV and thereby indirectly decreases aggressive behavior
- C. It indirectly improves both school performance and the child-parent relationship
- D. It increases aggressive behavior